

◦ Aim:

- (i) To study the hall effect and determine the Hall voltage (V_H) and Hall coefficient (R_H).
- (ii) To determine the type of majority carriers (i.e., to check if the semiconductor is p-type or n-type).
- (iii) To find the charge carrier density (n).

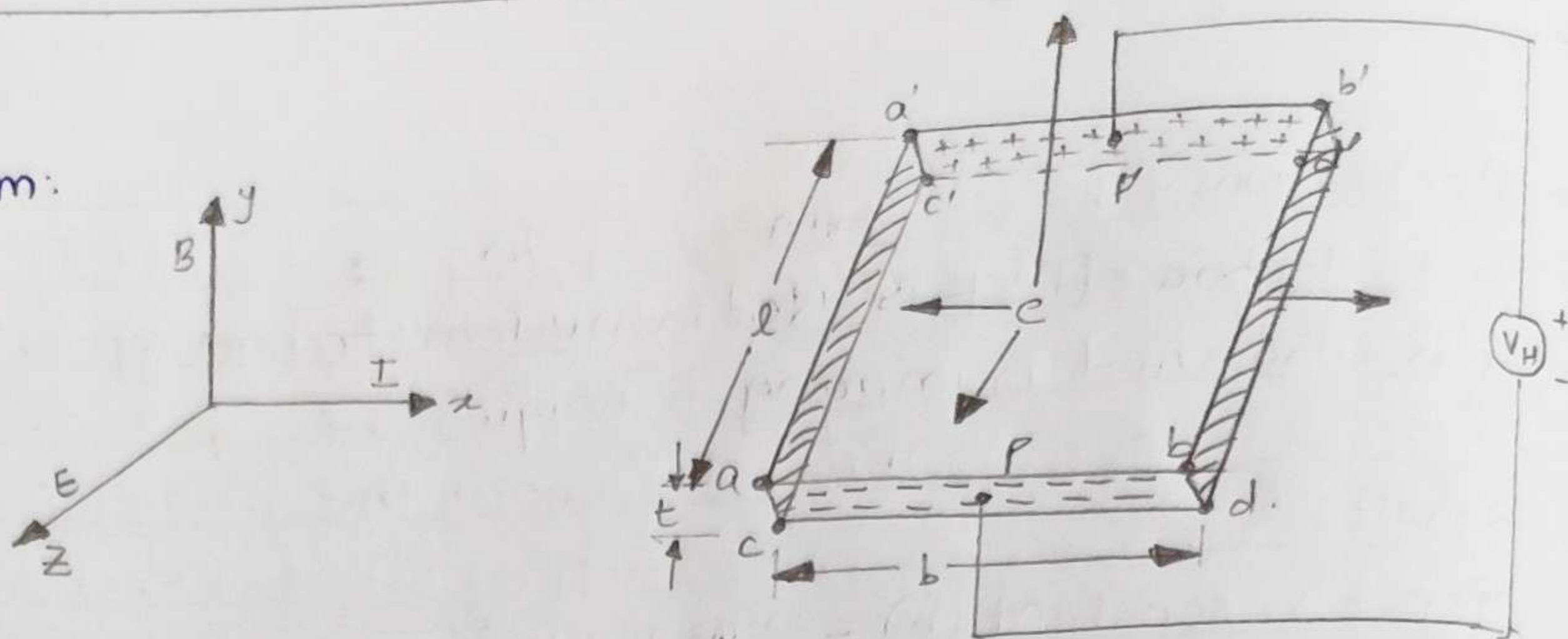
◦ Apparatus:

- Hall probe, Gauss meter, Power source, Multimeter, DC power supply, semiconductor crystal, voltmeter.

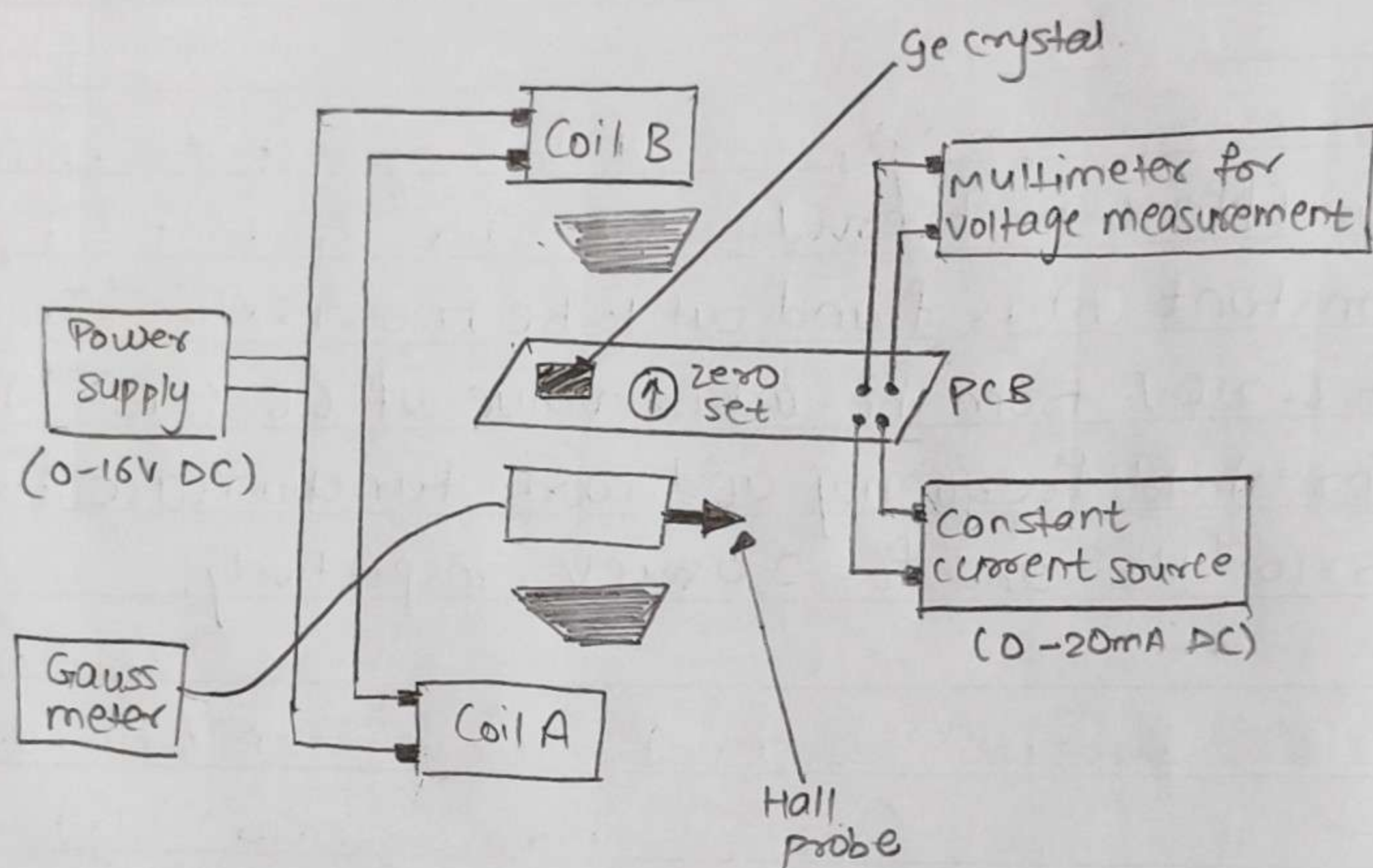
◦ Theory:

- In 1819, E.H. Hall observed that on placing a current carrying conductor perpendicular to a magnetic field, a voltage is observed perpendicular to both magnetic field & current. It was observed that the charge carriers, which were assumed to be electrons, experienced a sideways force opposite to expected direction.
- This phenomena is called as 'Hall Effect', and the induced voltage is called as 'Hall voltage (V_H)'.
- We know that a static magnetic field has no effect on charges unless they are in motion. When charges flow, a magnetic field directed perpendicular in direction of flow produces a mutually perpendicular force on the charge, consequently, electrons and holes gets separated and produce an electric field (E_H), setting up a potential difference, called 'Hall Potential (V_H)'.

Diagram:



[Figure (1)]



[Figure (2)]

o observations:

- $b = 0.5 \text{ mm}$

$B = 2.350 \text{ gauss Tesla.}$

Sr No.	Hall Voltage (when B and I in one direction)	Hall Voltage (when B and I in reverse dir.)	Current (I)	Mean V_H	$(\frac{V_H}{I})_{\text{mean}}$
1	0.1	0.1	0.1	0.1	1
2	0.4	0.3	0.35	0.30	1.167
3	0.6	0.6	0.6	0.48	1.250
4	1.0	0.8	0.9	0.62	1.451
5	1.3	1.2	1.25	0.80	1.5625
6	1.8	1.8	1.8	1.10	1.636
7	2.3	2.2	2.25	1.35	1.667
8	2.6	3.5 2.6	2.6	1.48	1.756
9	3.5	3.5	3.5	1.80	1.944
10	3.7	3.8	3.75	2.00	1.875
11	4.0	4.1	4.05	2.10	1.928
12	5.0	4.9	4.95	2.45	2.020

o calculations:

$$(i) V_H = \frac{0.10 + 0.35 + 0.60 + 0.90 + 1.25 + 1.80 + 2.25 + 2.60 + 3.50 + 3.75 + 4.05 + 4.95}{12} = \frac{126.1}{12}$$

$$\therefore V_H = 2.175 \text{ mV}$$

$$(ii) \left(\frac{V_H}{I} \right)_{\text{mean}} = \frac{1 + 1.167 + 1.250 + 1.451 + 1.5625 + 1.636 + 1.667 + 1.756 + 1.944 + 1.875 + 1.928 + 2.020}{12} = 1.5213 \text{ mV/A}$$

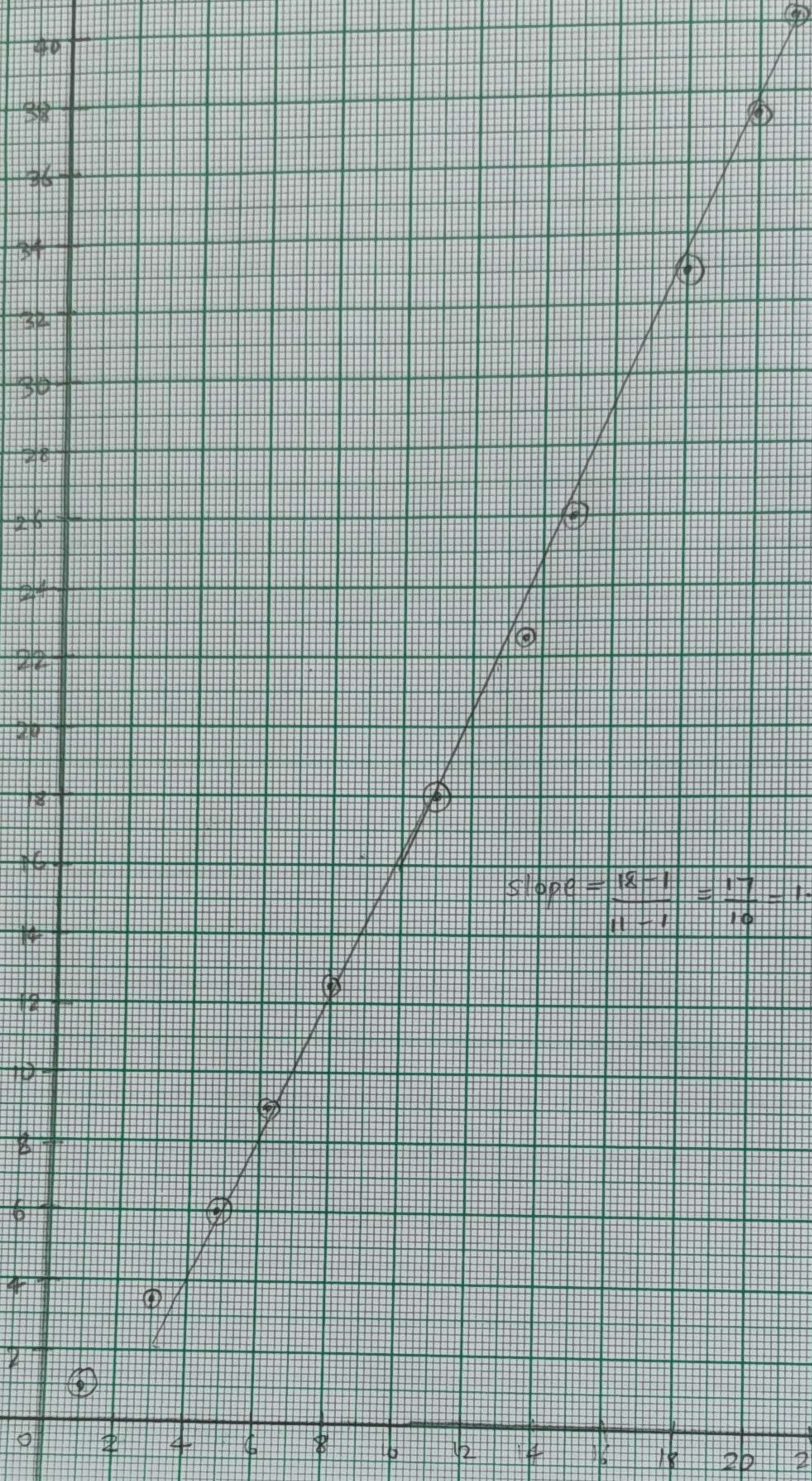
$$(iii) R_H = \left(\frac{V_H}{I} \right)_{\text{mean}} \times \frac{b}{B} = \frac{(1.5213 \times 10^{-3})(0.5 \times 10^{-3})}{2350} = 0.3236 \text{ m}^3/\text{C}$$

$$(iv) n = \frac{1}{R_{He}} = \frac{1}{(0.3236)(1.6 \times 10^{-19})} = 1.9313 \times 10^{19} / \text{m}^3$$

- Consider a semiconductor in the form of a flat strip. Let current I flow through the strip along x -axis. If a millivoltmeter is connected, no reading is shown. But, when a magnetic field is applied along the y -axis ($\perp$ to current), a deflection is produced. Refer to figure (1).
- If current is carried by electrons, $abcd$ is (-ve) charged and $a'b'c'd'$ is (+ve) charged. However, if current is due to holes, then $abcd$ becomes (+ve).
- At Thermal Equilibrium, Lorentz Force becomes equal to Electric Field.
 $\therefore eVB = eEH \Rightarrow VB = EH$
 where v is drift velocity $\Rightarrow v = \frac{I}{neA}$ where $A = \text{area} = bt$
 and E_H is electric field $\Rightarrow E_H = V_H/t$
 $\therefore \frac{IB}{ne(bt)} = \frac{V_H}{t} \Rightarrow \frac{IB}{neb} = V_H \Rightarrow \frac{IB}{ne} = V_H b$
 Hall coefficient, $R_H = \frac{1}{ne} \Rightarrow IBR_H = V_H b \Rightarrow R_H = \frac{V_H b}{IB}$
- o Procedure:
 - Connect the apparatus as shown.
 - Fix a value of magnetic field B using gauss meter & hall probe.
 - Slowly, increase the current and note down the voltages.
 - Now, reverse the current and again note the readings.
 - Find mean V_H and hence mean V_H/I .
 - Find Hall coefficient, $R_H = \left(\frac{V_H}{I}\right)_{\text{mean}} \times \frac{b}{B}$.

Y-axis
Label: V_{OH}
Unit: 10^{-3} mV

Scale: on X-axis, $1 \text{ cm} = 0.2 \text{ A}$
on Y-axis, $1 \text{ cm} = 0.2 \text{ mV}$



$$\text{slope} = \frac{18-1}{11-1} = \frac{17}{10} = 1.7$$

- Plot graph of V_H vs I and find slope to get R_H .
- If R_H is (+ve), semiconductor is p-type, otherwise n-type.
- Finally, find carrier charge density by $n = \frac{1}{R_H e}$.

° Precautions:

- Do not touch live wires.
- Make sure the circuit is connected properly.

° Applications:

- Hall effect is used in ignition systems & speed controls.

° Results:

- Hall voltage (V_H) = 2.175 mV
- Hall coefficient (R_H) = 0.3426 m³/C
⇒ semiconductor is of p-type
- charge carrier density (n) = $1.9313 \times 10^{19} / \text{m}^3$

° Conclusion:

- Hall effect is observed and graph between Hall voltage (V_H) and current (I) is plotted.
- semiconductor is found to be p-type
- Hall voltage, Hall coefficient, and charge carrier density are found out to be $V_H = 2.17 \text{ mV}$, $R_H = 0.3426 \text{ m}^3/\text{C}$, and $n = 1.9313 \times 10^{19} / \text{m}^3$ respectively.